

Bluestock Mutual Fund Analytics Capstone Project

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Executive Summary

The objective of this project was to perform end-to-end analytics on mutual fund industry data using Python, SQL, SQLite, and Power BI. The project involved data cleaning, database design, exploratory data analysis, performance analytics, advanced risk analytics, and dashboard development. The final solution provides actionable insights into mutual fund performance, investor behavior, SIP trends, and portfolio risk.

Project Objectives

- Build a clean and structured mutual fund database.
- Perform exploratory data analysis.
- Evaluate fund performance using financial metrics.
- Analyze investor behavior and SIP patterns.
- Develop a mutual fund recommendation system.
- Create interactive Power BI dashboards.
- Generate business insights and recommendations.

Data Sources

The project uses ten datasets:

1. Fund Master
2. NAV History
3. AUM by Fund House
4. Monthly SIP Inflows
5. Category Inflows
6. Industry Folio Count
7. Scheme Performance
8. Investor Transactions
9. Portfolio Holdings
10. Benchmark Indices

These datasets collectively provide information on mutual fund schemes, historical NAVs, investor activity, portfolio composition, and market benchmarks.

ETL Design

Extract

Raw CSV datasets were collected and imported into Python.

Transform

Data cleaning activities included:

- Date standardization
- Missing value handling
- Duplicate removal

- Validation checks
- Transaction type standardization
- NAV quality validation

Load

The cleaned datasets were loaded into a SQLite database using SQLAlchemy.

Database Design

The project implemented a star schema consisting of:

Dimension Tables

- dim_fund
- dim_date

Fact Tables

- fact_nav
- fact_transactions
- fact_performance
- fact_aum

This design improves query performance and reporting efficiency.

Exploratory Data Analysis

NAV Trend Analysis

Daily NAV trends were analyzed from 2022–2026 across all schemes. Strong growth was observed during the 2023 market rally while moderate corrections appeared during 2024.

SIP Inflow Analysis

Monthly SIP inflows displayed consistent growth throughout the study period, reaching record levels by the end of 2025.

Category Inflow Analysis

Equity-oriented categories attracted the highest net inflows compared to debt and hybrid funds.

Geographic Distribution

Certain states contributed significantly larger investment volumes, indicating strong regional concentration of mutual fund participation.

Investor Demographics

Middle-aged investors accounted for a substantial share of SIP investments.

Performance Analytics

CAGR Analysis

1-year and 3-year CAGR values were calculated for all schemes. Several funds demonstrated superior long-term wealth creation potential.

Sharpe Ratio

Sharpe Ratio was used to evaluate risk-adjusted returns. Top-ranked funds delivered strong returns while maintaining controlled volatility.

Sortino Ratio

Sortino Ratio highlighted downside-risk-adjusted performance and helped distinguish funds with better capital preservation characteristics.

Alpha and Beta

Alpha and Beta were computed against benchmark indices to measure excess return generation and market sensitivity.

Maximum Drawdown

Maximum Drawdown analysis quantified the worst historical decline experienced by each scheme.

Fund Scorecard

A composite scorecard was developed using:

- 30% Return Rank
- 25% Sharpe Ratio Rank
- 20% Alpha Rank
- 15% Expense Ratio Rank
- 10% Drawdown Rank

Advanced Analytics

Historical VaR and CVaR

Historical Value at Risk and Conditional Value at Risk were calculated to estimate downside risk exposure.

Rolling Sharpe Ratio

Rolling 90-day Sharpe Ratios were analyzed to evaluate changing risk-adjusted performance over time.

Investor Cohort Analysis

Investors were grouped by their first investment year to identify behavioral differences and investment patterns.

SIP Continuity Analysis

Investors with SIP gaps greater than 35 days were flagged as potential discontinuation risks.

Fund Recommendation System

A rule-based recommendation engine was developed using risk appetite categories:

- Low Risk
- Moderate Risk
- High Risk

Portfolio Concentration Analysis

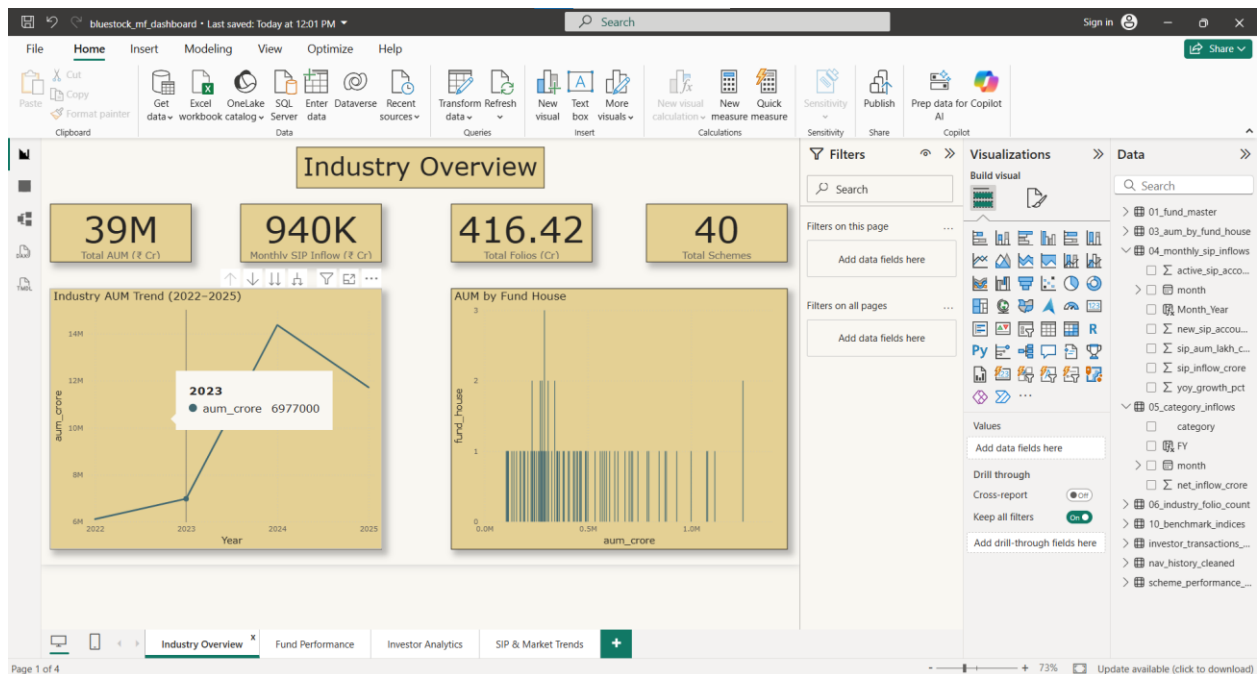
HHI concentration metrics were used to compare diversification levels across mutual fund portfolios.

Dashboard Development

A four-page interactive Power BI dashboard was developed.

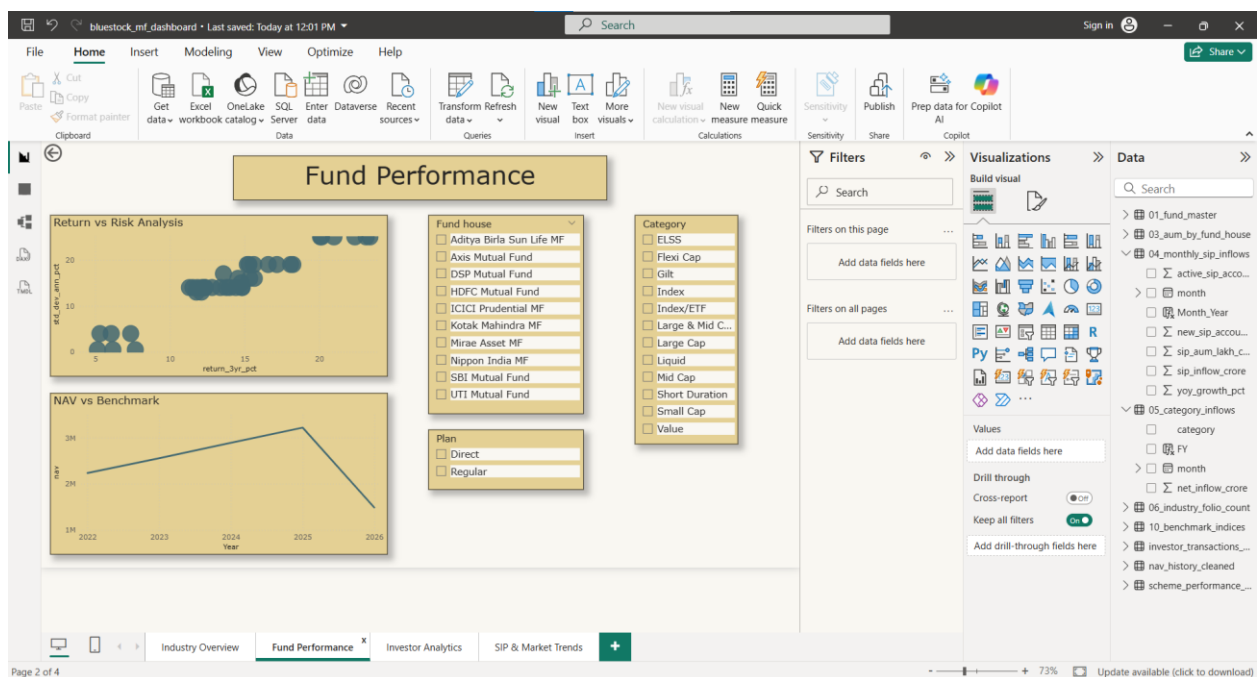
Page 1 – Industry Overview

- Total AUM
- SIP Inflows
- Folios
- Scheme Count
- Industry Trends



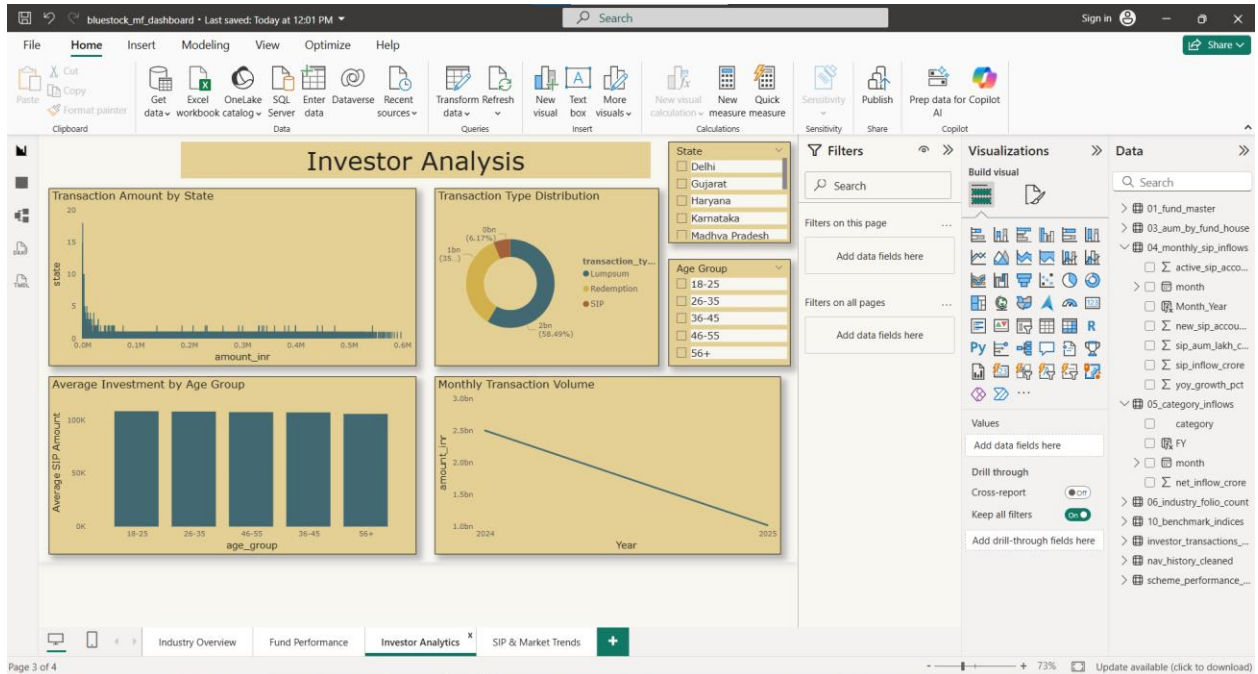
Page 2 – Fund Performance

- Return vs Risk
- Fund Scorecard
- Benchmark Comparison



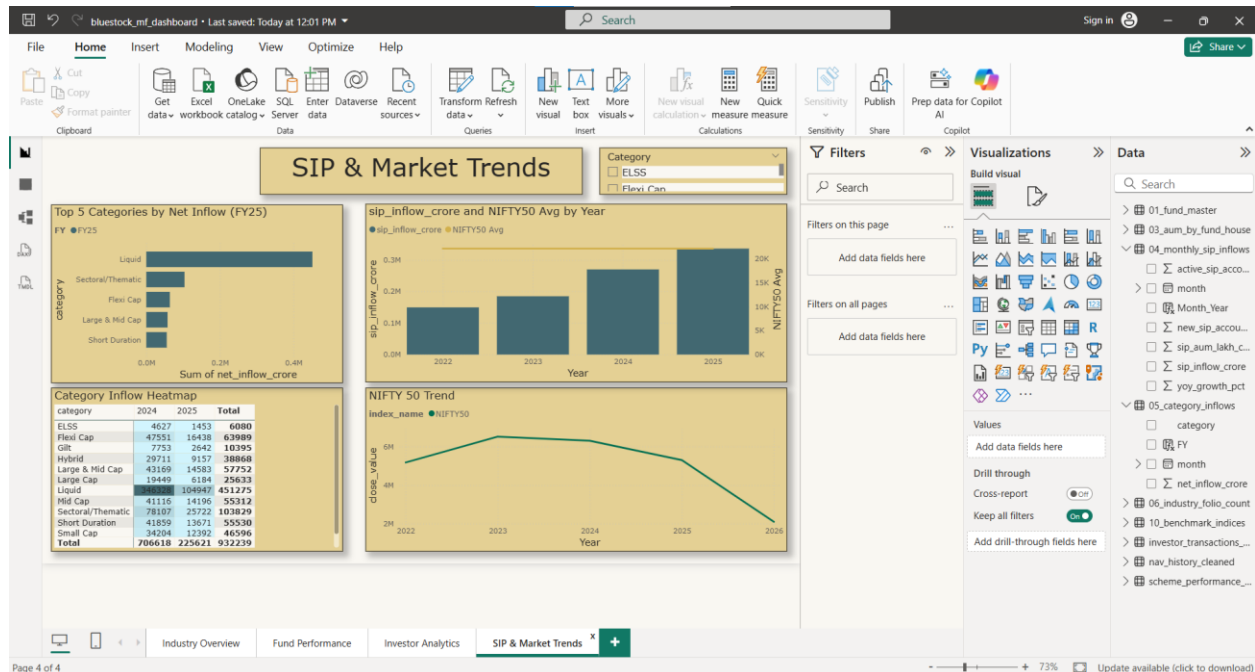
Page 3 – Investor Analytics

- State-wise Investments
- Transaction Type Distribution
- Investor Demographics



Page 4 – SIP and Market Trends

- SIP Growth
- Nifty Performance
- Category Inflows



Key Findings

1. Mutual fund industry assets experienced significant growth between 2022 and 2025.
2. SIP participation increased consistently throughout the analysis period.
3. Equity-oriented schemes attracted the highest investor interest.
4. Top-performing funds achieved superior risk-adjusted returns.
5. Portfolio concentration varied substantially across schemes.
6. Certain investor cohorts contributed significantly higher investment volumes.

7. SIP continuity remained strong for most investors.

Limitations

- Analysis is based on historical data.
- Future market conditions may differ.
- Simulated datasets may not fully represent live market behavior.
- Recommendation engine uses rule-based logic rather than machine learning.

Recommendations

- Investors should evaluate funds using risk-adjusted metrics rather than returns alone.
- Diversified portfolios generally provide lower concentration risk.
- Regular SIP contributions can improve long-term wealth accumulation.
- Fund selection should align with investor risk appetite and financial goals.

Conclusion

This project successfully demonstrated the complete analytics lifecycle including data engineering, exploratory analysis, financial performance measurement, advanced risk analytics, dashboard development, and business reporting. The final

solution provides meaningful insights that can support informed mutual fund investment decisions.

References

- [AMFI India](#)
- [NSE India](#)
- [Mutual Fund Industry Reports](#)
- [Python Documentation](#)
- [Power BI Documentation](#)
- [SQLite Documentation](#)